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SECTION 5: DRY LAND FARMING – IMPROVED PLANTED PASTURES AND FODDER CROPS

5.1 INTRODUCTION

Beef and milk production can be increased substantially throughout the year by planting improved pasture and fodder crop species. During the main growing period stocking rates of over 7 steers per ha can be achieved on some pastures. However, it must be stressed that to obtain the full benefit of improved crops, adequate fertilizer applications must be made, coupled with efficient general management. Where planted (sown) pastures will be utilized in a similar manner to veld pastures ie with minimal or no fertilizer and extensively rather than intensively managed, there will be little gain in planting a good veld pasture to an improved grass. However where the farmer intends to have a higher management standard and more intensive utilization, or where the veld pasture is poor, or the area is old reverted land, then planting to improved grasses will increase production.

Improved pastures have a part to play even on predominantly veld-grazed ranches. By planting as little as 50 hectares to improved species substantial improvement in production of hay, silage or improved standing hay will be made.Choice of species to use will depend largely on rainfall and soil conditions.

Several factors are important when selecting which grass or legume to use.

i) The overall dry matter production per unit area is one important factor, particularly for grasses. All the .grass species listed in are capable of moderate to high levels of production especially when fertilized with nitrogen, and will give several grazings (or cuts) per season.

ii)A second important factor is high feed values, particularly during the dry season. Those species that maintain a relatively high crude protein value into the dry season are more beneficial than those in which the crude protein level drops rapidly after flowering. Legumes are particularly valuable in this respect; whilst improved grasses tend to be highly better than veld grasses.

iii)Suitability to the local environment, especially soil conditions, is a third important factor. For example, some grasses and legumes are better adapted to water logged soils than others. This consideration will apply not only to dambos but to heavy upland soils which may become waterlogged during the rains.

iv)Method of use ie cutting or grazing is an important factor to be considered, particularly if forage harvesting machinery is not available. Some grasses eg Napier grass and Guinea grass are easily cut-and-carried manually whilst others e.g., Star grass (a creeper) are not suitable for this.

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Most of the grasses are suitable for conservation as hay; exceptions to this are Napier grass and Guinea grass. Most are suitable for conservation as silage. Molasses should be added to all with the exception of Napier grass, maize and sorghum.

Legumes show similar differences in their suitability for cut-and-carry and for hay or silage conservation. Leucena is very suitable for cut-and-carry.

v) The proposed duration of the pasture should also be taken into account. Some grasses, such as Rhodes, are only suitable for short-term leys' others, such as Guinea grass, are suitable for medium-term leys (up to six years). While grasses such as Star, Torpedo, Buffel and Bushman Mine Panic should be considered as permanent pastures.

The duration of perennial legumes in a pasture will depend largely on management.

vi) If mixed grass-legume pastures are considered, then compatibility of the species is of great importance. Aggressive, spreading grasses such as star do not generally mix well with legumes. Climbing legumes such as Siratro or Desmodium silverleaf should be used with all grasses.

5.2 GUIDELINES FOR ESTABLISHMENT OF SEEDED PASTURES AND FODDER CROPS

(i) Broadcasting
Broadcasting, by hand or fertilizer spreaders, is the easiest method of sowing pasture need. A more even distribution of the seed is obtained by mixing it with sand, saw-DUST OR FERTILIZER.

If the seed has been broadcast onto a prepared seedbed it should be covered lightly by harrowing, rolling with a Cambridge roller or by dragging a large branch over the soil.

ii) Drilling
This is a more efficient method for pasture establishment than broadcasting, as the seed can be placed at a desired depth and in closer proximity as the fertilizer. If weeding is necessary, sowing in rows makes this an easier operation. Small-seeded escapes and legumes should be sown at depths not exceeding 1cm. Larger seeded legumes, such as Leucaena, can be sown deeper (up to 3cm, deep). Sowing of pastures may be done with wheat drills but the coulters must be raised so that they barely touch the ground, otherwise the seed may be buried too deep.

The field should be rolled after drilling.

iii) Undersowing in maize
This is a useful method for establishment of pastures and some fodder crops. Seed can be either drilled or broadcast immediately after the first weeding, when the maize is about 30cm tall. Vegetatively propagated grasses such as Star can be established in this manner.

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Some advantages of undersowing are:
a) The maize crop will pay for the costs of the pasture establishment.

b) Undersowing immediately after the first weeding will reduce weed competition.

c) The maize crop will protect the young pasture plants and germinating seedlings from extremes of weather conditions such as high temperatures and rapid drying of the soil surface.

d) In mixed grass-legume swards, the shade from maize favours the legume component, resulting in a better grass/legume ratio.

iv) Direct seeding
This refers to the sowing of seed into a prepared seedbed without the presence of a cover crop such as maize. Where a mixed grass-legume pasture is to be established, two methods of seeding are available:

a) Sow the legume three to four weeks before the grass; this will allow the slower-growing legume to establish before the grass.

b) Sow the grass and legume in alternate pure strips (two to three metres wide) with the legume occupying one third of the total area. An extension of this is to sow the grass and legume separately in large pure blocks.

Seedbed preparation
Most species need a well-prepared seedbed; however, small seeds will need a finer seedbed than large seeds. The recommended procedure for general seedbed preparation is as follows:-

(a) Plough early in the dry season to kill off weeds and allow decomposition of crop residues;
(b) Further discings should be done during the dry season to breakdown large soil clods as much as possible;
(c) These should be followed by rolling to give a firm seedbed of moderately fine tilth. Seed to be dry-planted can be sown direct into this before the rains.
(d) Otherwise, a final cultivation should be done after the rains to kill off the initial weed crop and the pasture seed sown immediately.

i) Time of sowing
Grasses generally establish quickly from seed and need not be sown until mid-January.
Legumes establish more slowly and should be sown as early as possible
Dry planting, just before the rains are due, can be successful with both grasses and legumes.

However, inoculated legume seed must not be dry planted.

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ii) Seed rates

All seeded grasses and legumes mentioned in this handbook have recommended seeding rates however, the actual rate used may need to be changed depending on a number of factors.

a) Rates for drilled seed are usually lower than those for broadcast. It broadcasting, the recommended rates should be increased slightly.

b) If the grass is to be sown with a legume the normal seed rate of the grass should be reduced slightly

c) If the germination percentage of the seed is low the seed rate should be increased.

d) Where a mixture of legumes is sown, the rate of each legume can be reduced so that the total legume rate is about 4-6kg per ha.

Farmers are urged to establish their own seed nurseries until commercial seed production improves.

GRASSES MAINLY FOR PASTURE

i) Star grass and other similar grasses eg Torpedo grass, Kikuyu grass, *Panicum coloratum* cultivar. Bushman Mine Panic

Star grass will provide a permanent pasture, with-standing heavy grazing and high stocking rates. Torpedo grass is more suitable on very heavy soils particularly if prone to water-logging. Kikuyu will do well only in high rainfall areas, or under irrigation, where temperatures are moderate. Bushmen Mine Panic is very drought resistant and may remain green and leafy throughout the dry season. It performs well on most soils including heavy soils prone to some water–logging.

a)Establishment

Establishment of Star grass is from planting of rooted runners or from division of root stock.

Best results are obtained from hand planting at a spacing of 0.5 m x 0.5 m. Runners should be cut into lengths containing three to four rooted modes (joints) and folded together before planting. It is essential that the solid is pressed down well round the planted material.

Such a method is labour intensive and slow for planting large areas. Two alternative methods are:

i) Scatter runners at a high density over the area and disc into the soil, along with the fertilizer. A light rolling will be beneficial.

ii) Drop runners into a plough furrow, leaving some of the material protruding, and cover over.

Best results are obtained from using runners from the current season's growth. This may mean delaying planting until January.

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Preferably the land should be prepared initially by ploughing and harrowing; however Star grass can be established straight into veld that has been grazed or burnt down.

Planting should take place only when the rains are well established. When vigorious growth begins to be shown, now or lightly graze the pasture at regular intervals. This encourages lateral spread of runners and is essential to obtain rapid and even cover of the ground.

It will take usually at least one season to obtain a well established Star grass pasture. Some returns from the land during the first season may be obtained by planting the Star grass beneath a maize crop. The maize rows should be spaced at least 90cm apart.

Establishment of Torpedo grass is from planting of divided root stock and rhizomes (under-ground stems). It can quickly become a weed and is extremely difficult to eradicate. It should not be planted on, or close to, arable land.

Kikuyu grass is established from root stock or stem cuttings. Some cultivars are obtainable that can be established from seed (eg cultivar Whittet)

Bushman mine panic is established from root stock and runners. Material is available from the National irrigation research station (Mazabuka).

All these grasses can be established under maize.

a) Fertilizer

Establishment year – 400-600 kg single supers per ha and 150kg ammonium nirrate per ha should be applied and if possible disced into the soil.

Annual maintenance – 200-300kg single supers per ha should be broadcast at the start of the rains. Application of potassium is recommended only for established lands or where zero grazing and hay making are practiced. Soil and leaf analyses can be used as a guide in this.

150kg ammonium nitrate per ha should be broadcast at the start of the rains. For intensive management, nitrogen should continue to be applied at about one month intervals after commencement of grazing at rates of 120-170kg ammonium nitrate per ha per month. Urea is not recommended as field losses of nitrogen can be high.

Liming of the soil at a rate of 1-2 tones per ha may be beneficial if the soil pH below medium acidity (pH 4.6, $CaCl_2$ basis).

b) Management

It is essential that attention is given to the pasture during its establishment phase after planting. This means regular mowings or light grazings to encourage lateral runner

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growth. A light discing, once growth of runners is strong, will promote growth and destroy weeds.
After establishment is completed, Star may be grazed heavily and this is essential for good production and vigour. After several years the pasture may become sod-bound and growth reduced. It can be renovated by use of ripping tines and then application of fertilizer.

Intensively managed Star will maintain seven or more steers per hectare over the main growth period with average live weight gains of more than 0.7kg per steer per day.

If a hay crop is to be taken at the end of the rains, grazing should cease mid-late February and the pasture fertilized with nitrogen.

ii) Rhodes grass
Rhodes is tolerant to a wide range of soils, from light sandy soils to clays. It will not withstand water-logging however. It grows best on medium textured sandy clay loams. Rhodes provides excellent first-season grazing. Yields in subsequent seasons decline and generally re-planting is necessary after three or four years. It is drought resistant.

a) Establishment
Establishment is from seed. Sow in rows 0.5 m to 1.0 m apart, or broadcast, at a rate of 3-7 kg per hectare. Germination of fresh seed is improved if stored for 6 months before planting.

The seedbed should be prepared by ploughing and harrowing. Drilled seed should be placed no more than 1 cm deep and lightly rolled. Broadcast seed should be covered by a light harrowing and then lightly rolled. Sowing should not take place until the rains are well established. If germination is poor, stands should be allowed to flower and set seed before the first grazing is taken.

(b) **Fertilizer**
Establishment year – 400-600 kg single supers per ha plus 150kg ammonium nitrate per ha disced into the soil.

Annual maintenance – 200-300kg single supers per ha with 150kg ammonium nitrate per ha broadcast at the start of the rains.

For intensively managed pasture, continue to apply nitrogen at monthly intervals during the growing season at rates of 120-170kg ammonium nitrate per ha per month.

If Rhodes is grown as a ley pasture within an arable crop rotation, apply 200-300kg 'D' compound per ha in the last one or two years of the ley instead of single super and ammonium nitrate.

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(c) Management
New stands should be allowed to set seed before the first grazing particularly if establishment is poor.

Rotational grazing is desirable for Rhodes; under continuous grazing the pasture thins out and quickly becomes weedy.

Peak grass production is obtained usually during the first and second seasons. Animal production is sometimes lower than in Star grass. However, palatability is better and establishment easier. Stands which show degeneration sometimes may be renovated by allowing to flower and set seed and new seedlings given time to establish themselves. However it is usually more satisfactory to plough and re plant.

Rhodes is an ideal grass to use as a temporary ley in rotation with arable crops. Rhodes is tolerant of burning; this may be used as a tool to clear old growth and weeds.

iii) Buffel grass
Buffel is tolerant of a wide range of soil conditions (except water-logged soils). However it finds its best application on light soils in the medium and low rainfall areas. It is very drought resistant and will remain green well into the dry season. It is very adaptable and can make good hay. It is suitable only as a permanent pasture.

a) Establishment
Establishment is as for Rhodes grass, sown at a rate of 3-4 kg seed per hectare. If a drill is to be used, the seed should be passed through a hammer-mill, to remove the bristles. Fresh seed has a dormancy period of 9-12 months and should not be used.

b)Fertilizer
As for Rhodes grass

c) Management
Buffel persists well under close grazing, either rotational or continuous. It makes good hay and is tolerant to fire; burning can be used to clear old growth and weeds at the end of the dry season. However, old dry stems often sprout green shoots during the dry season and these can contribute to dry season grazing. Topping of stems immediately after the wet-season grazing is not recommended.

iv) Guinea grass
Guinea grass prefers medium to light-textured soils and does not tolerate heavy clay and soils that are water-logged or liable to prolonged flooding. It is moderately drought resistant.

The cultivars Hamil, Ntschisi and Makueni are adapted to the high rainfall area above 1000mm per year) whilst the cultivars Gatton and Petrie (Green Panic) are suited for the medium rainfall areas. Guinea grasses are remarkably shade tolerant. Herbage yield is higher than most other grasses and palatability is good.

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a) Establishment
Guinea grass can be established by seed or by tuft splits. Fresh seed should not be sown. Establishment is similar to Rhodes grass, at a seed rate of 37 kg per hectare. Seed can be sown slightly deeper than most other grasses. Emergence is slow

b) Fertilizer
As for Rhodes grass, Guinea grass responds well to nitrogen and, if grown, should be intensively managed.

c) Management
Guinea grass can be heavily stocked but plants should not be grazed below 15-20cm from the ground. Rotational grazing is recommended with at least three weeks resting period between grazings. Guinea grass can also be cut and fed green or made into hay. It is tolerant to fire.

Guinea grass responds well to irrigation during the hot dry months (September to November).

GRASSES MAINLY FOR FODDER PRODUCTION

i) Napier (elephant) grass
This is a to all, deeprooted grass similar in appearance to sugarcane. It is one of the highest yielding grasses and does well under irrigation. It is generally cut for silage or green feed, but can be grazed. It is drought resistant and moderately frost resistant.

a) Establishment
Establishment is from stem cuttings or root splits. Old, woody stems do not make good planting material and at the start of the rains, better results are obtained from using root splits. Later in the season, cuttings from new stems can be used.

Stem cuttings should contain three to four nodes (joints). Buds situated at these nodes initiate growth. Cuttings should be placed at an angle in the ground such that two to three nodes are buried in the soil and one or two remain uncovered. The basal end of the cutting should be buried ie the buds should be positioned above the nodes, rather than below.

Plant in rows 0.5 m apart; the wider spicing should be used in low rainfall areas. Within the row, cuttings or root splits should be spaced 0.3 m to 0.9 m apart:

Land should be prepared initially by ploughing and harrowing. If planted direct into veld, then the veld vegetation should be grazed or burnt down first. Do not plant on land subject to flooding or water logging.

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Annual legumes such as cowpea, lab-lab bean or velvet been can be planted between the rows. These will improve the soil, suppress weeds and provide additional high quality fodder. If legumes are interplanted, Napier should be in rows 1.5m to 2.0m apart.

Napier generally requires re-planting every three to four years.

(b) Fertilizer
Establishment year – 500kg 'D' compound per ha disced into the soil if possible.

Annual maintenance – 200-300kg 'D' compound per ha at the start of the rains. For sustained high yields apply 120kg ammonium nitrate per ha after each harvest.

(c) Management and utilisation
Cut the Napier grass when it has reached a height of about 1m. Do not allow to grow taller before harvesting as stems and leaves become coarse and unpalatable. Harvest the crop at a height of 20cm from the ground. Do not cut lower as re-growth will be delayed and reduced.

Before feeding or making into silage, stems and leaves should be chopped into short lengths, Napier makes good silage without further additives being required, a short wilting period, if practical, is generally desirable.

Grazing is not recommended generally as animals can become injured easily by old cut stems and the sharp mid-rib points of broken leaves. Excessive treading will delay recovery. In practice, grazing will shorten the life of the lay.

During the rains harvesting can be done at about 4-week intervals. Growth may continue during the dry season depending on soil moisture and temperature. This growth will be improved if the crop is allowed to flower before the last wet season cut.

Combined with perennial climbing legumes such as siratro or glycine, Napier provides a good resting ley pasture.

ii) Maize
Maize will grow on most soils except those prone to water-logging. However, yields on sandy soils are likely to be relatively low. Generally, fresh fodder yields range from 20-80 tones per hectare when the whole plant is utilized. A quick, but approximate, method of judging fresh fodder yield is to estimate the number of bags of grain likely to be obtained per hectare; this number is then equivalent to an equal amount of tonnes of fresh fodder per hectare. For example, a grain yield of 50 bags per hectare will yield approximately 50 tones of fodder.

Maize for fodder is generally ensiled (made into silage) and it is perhaps the best crop for making silage. It can also be fed green or grazed. The stover of maize grown for grain can be used for dry-season grazing.

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a) Establishment
Establishment of maize is simple and germination and growth of seedlings is fast. With irrigation, planting in most months of the year is possible. However, soil temperature should not be below 15°C.

The seedbed should be free of weeds but a fine tilth is not necessary. Depth of sowing should be between 5-10cm at a seed rate of 15-20kg/ha. Rows should be 74-90cm apart. On well fertilized soils, herbage yields can be increased by planting denser stands but the proportion of cobs is reduced. If maize is grown for grazing, closer rows should be used.

Weed control is important. This can be done by hand or mechanical means, with or without the use of herbicides. Recommended herbicides include Gasaprim and primagram for control of a wide range of weeds, Eradicane for control of grass weeds and suppression of nut-sedge and Basagran for spot spraying on nut-sedge patches. Maize advisory leaflets should be consulted for detailed information.

Pests:
Stalk borer should be controlled by spraying Carbaryl at 2kg/ha, Endosulphan at 2l/ha, Endosulphan at 2 kg/ha. The crop should be examined frequently after 30 cm high and sprayed as soon as caterpillar damage is noted.

b) Fertilizer
Apply 300kg 'D' compound per ha plus 300 kg urea/ha disced into the soil.

c) Management and utilization
Harvesting for ensilage is done generally at the milk or dough-ripe stages of grain maturity. For this, the whole plant is harvested and chopped either by machine or hand.

Provided the maize is well-compressed, good silage results almost without fail. Maize silage provides excellent fodder for fattening cattle and milk cows. It should be supplemented, however, in these cases with a high-protein feed.

Alternatively, the cobs may be harvested for commercial use and if taken off as green maize, the remaining plants made into silage, cut for silage or grazed as green stover.

If dried stover is grazed during the dry season, a useful practice is to plant the borders of the field with perennial legumes such as Siratro (low to high rainfall areas), Glycine and Desmodium Greenleaf (medium to high rainfall areas) or Desmodium silverleaf (high rainfall areas). Legumes will provide a highly desirable protein supplement.

iii) Sorghum
Fodder sorghums can be annual or short-lived perennials, ranging from 1m to 3m in eight. Drought and heat resistance are good: silage crops can be produced with as little 550 mm of rainfall. Feed value compares very favourably with maize.

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a) Establishment
Sorghum will grow on a wide range of soils provided they are well drained. Sandy loams, or heavier soils, are preferred.

Land should be prepared by ploughing and harrowing. Seed should be drilled in rows about 45cm apart, with plant spacing within the row of about 7cm. Seed rate should be about 15 kg per ha. Sow at a depth of 3-5cm. Seed should be treated with Captasan – M or other similar dressing to prevent pest attack.

b) Fertilizer
Apply 200-300kg 'D' compound per ha as Basal dressing, plus 150kg ammonium nitrate, disced into the soil.

c) Management and utilization
Weed control at the early stages of growth is essential. This can be done by shallow cultivation between the rows or by use of herbicides. Recommended herbicides are similar to maize.

Pests such as cutworm, pink borer and stalk borer should be controlled as recommended for stalk borer under maize. Spraying against pink borer should be standard practice commencing as soon as germination is complete and repeating 14 days later. For cut worm and pink borer, the chemical should be applied to the base of the plant and adjacent soil.

Sorghum grown for forage may be utilized as silage or hay (whole plant), stover or as a soiling crop (cut and fed green). Grain heads may be fed direct or thrashed and the grain used mixed with other feed. Sorghum cut for silage or hay should be fairly mature, that is, when the seeds are at, or near, the hard dough stage. If cut too early, there is less dry matter, palatability is lowered, prussic acid content is higher (drying as hay, however, destroys this acid) and high moisture content of immature sorghum may result in poor silage.

If sorghum is grown for soilage, several successive sowings should be made in order to extend the soiling season. However, re-growth will occur after cutting, special care must be taken to avoid bloat and prussic – acid poisoning from sorghum that is fed immature and green. The sorghum can be cut any time after it is 0.6m high but early cuttings or small second-growth stalks will contain dangerous amounts of prussic-acid and feeding of this young growth is not recommended. Sorghum grown in drought conditions is also usually high in acid content.

Sorghum silage and well- cured hay is generally safe to feed to animals.

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PASTURE AND FODDER LEGUMES

Legumes grown as pure crops provide high-protein supplement feeds, mainly for use during the dry season. As they are low in energy requirements they should not be used as sole diet; fresh grass, hay or silage should be provided also.

1) Leucaena

Leucaena is a leguminous shrub or small tree that produces a high protein fodder. Most of Zambia lies within the climatic limits required for its growth but at altitudes above 1200 m growth will be reduced.

Maximum growth occurs during the rains but growth may continue during winter at a reduced rate. Leucaena is extremely drought resistant and will retain green leaf throughout the dry season in most areas.

Initial growth of seedlings is slow and establishment may take a full season. Leucaena is not affected by occasional light frosts. Severe frost will cause loss of leaf but the plant is unlikely to be killed.

Yields of 2.5-4.0 tonnes/ha dry matter (7.5-12 tonnes fresh material) can be expected from a closely spaced crop of established leucaena over a 6 week growth period during the rains. In winter with irrigation, up to 2 tonnes/ha dry matter (7 tonnes fresh material) can be obtained over a 10week growth period. These yields include both leaf and small branches up to 5mm in diameter.

Leucaena has a high protein value (28-31 percent CP for dried leaflets) and carotene content can be twice as high as for Lucerne.

Several cultivars are available now in Zambia including Peru, Hawaiian Giant and Cunningham. Seed production is prolific and hand harvesting simple.

Seed is available from the seeds section, Mount Makulu Research Station, Chilanga.

Leucaena will grow on wide range of soils although neutral or slightly basic soils are preferred. It will tolerate moderately acidic soils (very acidic soils should be limed).

Established plants will tolerate water-logged conditions but young plants are extremely susceptible to poor drainage.

a) Establishment

If Leucaena is to be grown in closely spaced stands, prepare an adequate seedbed by ploughing and harrowing. The seed bed should be as free of weeds as possible as young leucaena seedlings are extremely susceptible to weed competition.

Ridging is desirable if the crop is to be established on land likely to be temporarily water-logged during the rains or if water conservation is important.

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1) Scarification
Seed must be scarified before sowing by placing in very hot water (80° C for 2 minutes) and then drying. Boiling water must not be used. When only a small number of seeds are to be planted, the broad end of the seed can be rubbed on sand paper or rough stone to break the seed coat.

2) Inoculation
It is essential that seed be inoculated with the correct rhizobial inoculum before sowing, otherwise growth will be poor. Alternatively soil from beneath leucaena trees that are known to be nodulating can be used. This method, however, is not so reliable.

(3) Sowing
Seed should be sown at a depth of 2 cm as soon as the rains have become established. Emergence will take 10-14 days.

Spacing and seed rate will depend on whether the crop is to be used for cutting or grazing. (See Table below)

Table 5.1: Recommended spacing and seed rates for leucaena

Utilisation	Seed rate (kg/ha)	Spacing in crop	
		Between rows (m)	Within rows (cm)
Forage harvester	30	0.5 - 0.75	10
Hand cutting	17 - 30	1.0	15 – 30
Grazing	11 - 22	3.0	10

Maize drills can be used for direct sowing of leucaena seed. The drill should be set to sow about 40 seeds per metre.

If irrigation is available, seed can be sown before the rains. However, do not sow before early September. Sufficient water should be applied to keep the soil around the seed moist.

Leucaena can also be established by transplanting 1-2 month old seedlings, or when the plants are about 15cm high. This method is laborious but ensures good establishment. Its use is recommended only under special circumstances e.g., establishment of a seed orchard or when small areas are to be established near dairies or small farms. Transplanting also eliminates the need for inexperienced farmers to handle inoculum. (if the seedlings have been provided by an established nursery with trained personnel).

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Seedling nurseries should be established in late October so that the young plants are ready for transplanting when the rains arrive.

Place inoculated seed in plastic planting sleeves containing top soil mixed with a little fertilizer (single super phosphate): 2 seeds per sleeve and thin later to one seedling. Water daily. Transplant when 1-2 months old, using all the soil. Clear weeds away from around the planting hole.

b) Fertilizer
400 kg per ha single supers are recommended as a basal application at planting and a further 200 kg per ha D compound as an annual top dressing. 400 kg per ha single supers are equivalent to about 40g per metre- row. Two normal sized match boxes, when filled with single supers, hold about 40 gm

c) Weed and pest control
Young leucaena seedlings are very susceptible to weed competition. They should be kept free of weeds for the first three months after emergence.

For chemical weed control, Dacthal (chlorthal) at 8kg/ha and Treflan (trifluralin) at 1.1 kg/ha have been used successfully. Either of these chemicals should be applied pre-planting and Treflan incorporated into the soil immediately after application.

Termites can be a problem with young plants. Control by use of dieldrin dissolved in water. Do not mix with seed as inoculum may be killed by the insecticide.

No major diseases have been reported for leucaena

d) Utilisation
Leucaena can be cut and fed green or as dried fodder; it can also be grazed as a browse shrub. There could also be a market for dried, milled leucaena as a component of animal feeds.

i) Cutting
Cutting can be done either by hand or by forage harvester.

Hand cutting: -cut main stems back to a height of about 30cm after flowering of the newly planted crop has occurred. Cut re-growth when branches reach a length of 1m. Do not allow growth to exceed this length because then, the proportion of wood to leafy material becomes too great. Cattle will generally take stems measuring up to 5mm in diameter.

Forage Harvester: -take the first cut when flowering starts and subsequent harvests when the branches are about 1m in length. The height of cut should be about 15cm. Spacing between rows should be adjusted to fit equipment wheels.

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ii) Grazing
Leucaena can be grazed as a pure crop or when interplanted with grass pasture.

If grazed as a pure crop, plant in rows about 2m apart. Access to such a crop must be strictly limited (3 days in 7 are used in Australia and the Philippines), as excessive grazing can lead to poisoning of the animals(has an anti nutritional factor called Mimosine). The area must be well fenced. This is a suitable system for dry-season feeding: the leucaena can be divided into fenced blocks that are opened up successively.

Leucaena used as part of a grazed pasture should be planted if possible the season before the grass is due to be planted. This will enable the leucaena to establish well and without competition. The area planted to leucaena should not constitute more than one quarter of the total pasture area; the rows should be at least 3m apart.

Leucaena can be incorporated into already established pasture by ploughing and planting strips: cattle would have to be excluded for the first season. Transplant seedlings could also be used in strip planting. Main stems in grazed or browsed leucaena should be cut back to a height of about 2m after first flowering. On established shrubs, cattle rarely eat stems more than 5mm thick so there is little danger of over-grazing.

e) Drying
When leucaena is cut and carried to the animal it can be fed green or dried. Fresh leucaena is generally readily accepted by cattle but wilted leucaena may not be. Dried leucaena may be mixed or fed separately.

Natural sun drying may be achieved by laying branches on racks or plastic sheeting; or by stacking them against their own stubble. After three or four days of fine weather the leaves will be dry enough to be beaten or stripped off into sacks. If dried by stacking branches against the stubble, then a wheeled container can be moved along the row and the dried leaf stripped into it. If the leaf is dry enough to be crushed easily it will store without deterioration. A (50kg) fertilizer bag holds about 17kg dried leaf.

Dried leaf can be milled in a maize hammer mill: a fertilizer bag will hold about 17kg milled leaf. This is easier to mix with other feeds.

Leucaena can also be dried by artificial drying units.

(f) Nutritive value
Ungraded leaf (mid-ribs, pods, flowerheads and small twigs less than 2mm diameter as well as leaf) has a crude protein content of 20-25 percent. Graded leaf has a CP content of 28-31 percent. Ungraded leaf plus stems up to 5mm diameter have a CP content of 16-20 percent. Stems alone have up to 9 percent CP and also have some value as roughage.

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g) Feeding
When fattening cattle or when providing supplementary feed to dairy animals, no more than 6kg fresh leaves or 10kg leaves plus stems should be fed per head per day. Dried leaves can be fed at a level of 2-2.5 kg per head per day. Dairy animals should be fed either at milking or immediately after. Feeding 2 hr prior to milking may taint the milk. Pasteurisation, however, removes the taint.

One fertilizer bag of unmilled leaf will be sufficient for one animal for 3 days. One maize bag will be sufficient for 8 days.

One meter-row of leucaena will produce approximately 1.0-1.5kg fresh leaf plus small stems over a 6 week growth period during the rains.

Poultry rations should not contain more than 5 percent dried leaf.

Rations for fattening pigs should not contain more than 20 percent dried leuceaena leaf..

h) Toxicity
Leucaena contains an amino-acid known as mimosine which can reduce growth rates, cause loss of hair and reproductive problems when fed in large quantities over a period of time (ie 6 months). Non-ruminants are more likely to suffer than ruminants. It is important that feeding be strictly controlled: leucaena should be used only as a high protein supplement fed in small amounts. When used in this manner it is a valuable and useful feed.

2) Lablab (*Dolichos*) bean

This is a large-seeded vigorous annual (sometimes biennial) legume. It makes excellent growth during the rains and shows good drought resistance. This enables it to be grown in low rainfall areas. Lablab will grow on a wide range of soils but will not tolerate water –logging.Leaf-shed (loss of leaf material) may occur during the dry season but under irrigation it will continue to remain productive particularly during August-November. It is not recommended for high rainfall areas where leaf diseases may cause problems.

Lablab withstands close cutting or grazing better than cowpea. Herbage yields are high crude protein value is good (25 percent).

The recommended cultivars are cv. Rongai and Selection No. 3 (mid-March). Rongai is later-flowering (mid-May) than Selection No. 3 (mid-March).

a) Establishment
Prepare a smooth seedbed by ploughing and harrowing. Broadcast fertilizer and disc in.

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Sow seed at a rate of 10-15 kg per hectare, either broadcast or in rows 50-100 cm apart. Sow when the rains are established; or under irrigation, from late August. Recommended planting depth is 2-3 cm.

i) Scarification
Seed should be scarified by placing in very hot water (80°C) for two minutes.

ii) Inoculation
Inoculation of seed sown in Zambia soils is not required generally.

b) Fertilizer
A basal application of 300-400kg single supers per ha should be applied and incorporated into the soil. If satisfactory nodulation of the roots has occurred no nitrogen will be required. If nitrogen deficiency is apparent, apply 150kg per ha ammonium nitrate.

c) Management and utilization
Lablab can be grazed or cut. It does not make very satisfactory hay as the thick stems can be slow to dry, resulting in leaf shed. Bloat seldom occurs and is not considered a problem.

The first grazing or cutting will be available 6-8 weeks after emergence. Subsequent harvestings will be available at 4-6 week intervals. Do not cut or graze lower than 15 cm from the ground. Milk production is generally higher than from grasses. The milk may be tainted but this disappears on pasteurization.

It does not make satisfactory silage unless molasses or maize is added.

3) Siratro, Stylo and other pasture legumes

A number of legumes can be established as pure pastures and as such have useful functions in providing grazing through much of the dry season. Pasture legumes retain their greenness longer and maintain a higher crude protein level during the dry season than grasses. Table 5.2 below gives recommended legumes for dry season pasture grazing Such pastures are useful in long-term crop rotations (ie., at least five years) as they will improve soil fertility.

a) Establishment
Faster and more effective establishment is obtained if the legumes are planted into clean, well prepared seed beds. However, establishment can also be made into veld where grasses and weeds have been grazed down or burnt.

Broadcast (either in strips or over the whole area) or sow in rows, at seed rates of 4-8kg per ha. The higher rates are commended for low rainfall areas. Broadcasting is preferable to sowing in rows, as covering of the ground will be achieved more rapidly. If possible, apply a mixture of legumes to take advantage of different characteristics such as

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growth habit, frost, fire and drought tolerance. The seeds should be lightly covered by soil, if possible, by light harrowing or by dragging branches over the soil surface.

If the legume seeds do not require inoculation (see below) then the seed may be applied just before the rains are due. In this case, about 50 percent of the seed should be left unscarified (if this is required) to allow for isolated storms before the main rains are established. Inoculated seed should not be sown until the rains have arrived; sowing then should take place as early as possible.

Scarification
Some legume seed will require scarification before planting. Consultation is required to determine which legumes require this.

Inoculation
Some of the legume seed may require inoculation before planting. Appendix D2 should be consulted to determine which legumes require this.

b) Fertilizer
At establishment, a basal application of 400-600kg ha single super phosphate should be made and disced into the soil if possible. In subsequent years, maintenance applications of 200kg per ha single supers should be made annually or once every two years. No nitrogen applications will be required.

c) Management
No grazing, or only light grazing should be allowed to keep down weeds and grasses during the rainy season of the establishment year. Flowering and seed set should be allowed during the establishment year and, if necessary, during the second year.

In general, pure legume pastures should be managed for grazing during the dry season only; grazing should not be permitted during the rains. In this way a large bulk of material is made available for the animals during the dry season. Once established, pastures can be fairly heavily grazed down during the dry season.

Once a legume area has been established, cattle can be used to spread the seed and so assist in establishment of other areas. Graze the cattle for short periods when the legume is seeding, then transfer to the new areas, keeping them there for at least four days. Some seed will pass through undamaged and be deposited in the animal dung. This is best done just before the rains are due as dung seed often germinates rapidly.

In general, pasture legumes are not very fire-resistant and every effort should be made to prevent burning of the pasture.

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Table 5.2: LEGUME SPECIES FOR DOMINANT-LEGUME PASTURE DRY SEASON GRAZING (x = Recommended)

Soil Texture =>	Rainfall								
	Low			Medium			High		
	Light	medium	heavy	Light	medium	heavy	Light	medium	heavy
Archer dolichos	-	-	-	x	x	x	x	x	-
Centro	-	-	-	-	-	-	x	x	x
Desmodium Greenleaf	-	-	-	x	x	x	x	x	x
Desmodium silverleaf	-	-	-	-	x	x	x	x	x
Glycine	-	-	-	-	-	-	-	x	x
Siratro	x	x	x	x	x	x	x	x	x
Stylo	x	x	x	x	x	x	x	x	x
Townsville Stylo	x	x	-	x	x	-	x	x	-

4) Cowpea
This is a bushy, trailing or climbing annual legume. It is used as a greed fodder, hay or silage. Spreading, indeterminate varieties should be used for fodder production. It can be grown mixed with maize, sorghum or Napier grass for ensilage.

5) Soybean
This is a large-seeded, bushy annual legume. It is used as a pasture and green fodder, crop; it can also be made into hay or silage.

Varieties specifically bred for fodder should be used preferably. The crop is harvested for fodder usually at the developing pod stage but before leaf shed commences. If made into hay, the crop is dried in windows and attached. Yields of about 20 tons green matter per ha can be obtained. The protein content of fresh herbage is about 15 percent. Soybean for fodder can be grown mixed with maize or sorghum.

6) Velvet beans
This is a twining annual legume. It is usually grown for green fodder or for ensilage with maize or sorghum. It is difficult to make good hay with velvet bean as frequent turning is required and many pods become detached. If grown fro hay or silage, the crop should be cut when the first pods reach full size but are still green. As with most legumes, additional molasses or maize is required to order to make good silage out of velvet bean.

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When grown with maize as a fodder mixture for ensilage, velvet bean is usually sown at the same time of planting either in the same rows or between the rows. If grown for improvement of stover grazing, then velvet bean should be sown after the last weeding and between the maize rows.

Seed rates for fodder production are generally lower than for grain production and 15-20kg per ha can be used.

GRASS-LEGUME MIXTURES

i) Introduction

In grass-legume mixtures, the grass usually forms the main bulk of the herbage and the legume increases the bulk and raises the crude protein value of the total herbage. Sometimes the bulk of herbage is not increased, particularly if the grass is very high – yielding and vigorous in growth, but invariably the presence of legumes will improve the quality.

Generally, in grasses mixed with tropical perennial legumes there is only slight transfer of nitrogen from the legume to the grass in the short term; farmers should not expect major nitrogen contributions from the legume component of the pasture (this is in contrast to temperature grass-legume mixtures). The main contribution of the legume will be to increase the bulk of herbage and raise the crude protein content of the grass pasture.

Mixed grass-legume pastures make better quality hay than the pure grass stands and are easier to make than for the pure legume. This is because the legume, not being so thick, dries faster and with less leaf loss.

In situations where the farmer is unable or unwilling to apply high levels of fertilizer nitrogen to his pasture, grass-legume mixtures will be more productive per hectare than the pure grass. However an improved grass pasture, well fertilized with nitrogen, will be more productive, and easier to manage, than the grass-legume mixture. The economics, of course, of heavily fertilizing a grass pasture with nitrogen will depend on the farmers' particular circumstances and the current cost of nitrogen.

To maintain successful mixed pastures (ie with the legume component comprising between 20-50 percent of the pasture) careful management is required. Particular care should be taken in the early stages and up to two years may be required for full establishment and spread of the legumes. Incorrect management either will revert the pasture to pure grass, thus losing the bulk and quality contribution of the legume, or lead to a dominance of the legume which for general grazing during the wet season will provide an unbalanced diet for the animal (dominant-legume pastures provide a good supplement for dry-season grazing).

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ii) Choice of legumes

Not all legumes are compatible with a particular grass and some care should be exercised in the choice of mixture both in the suitability of the legume to the grass and the legume to the environment in which it will grow.

Star grass is not recognized generally as mixing well with legumes due to its spreading, competitive character. However with care, training and climbing legumes such as glycine, sitratro and Desmodium silverleaf can be maintained. Leucaena, grown as a browsed shrub within the pasture, will be above the star grass award and will be no problem to maintain.

Rhodes grass, Bufffel grass and Guinea grass generally combine well with legumes: siratro, glycine, stylo, Lucerne, Desmodium silverleaf, Desmodium green leaf and Leucaena all can be maintained satisfactory.

Kikuyu grass does not combine well with legumes because of the grasses' strong and aggressive growth characteristics.

Panium coloratum (cultivar Bushman mine panic) is not an aggressive grass and legumes such as siratro, glycine and Desmodium can form stable associations with it.

iii) Legume seed rates

Siratro	4-6kg per hectare
Stylo	4-6kg per hectare
Glycine	5-7kg per hectare
Lucerne	6-8kg per hectare
Desmodium Greenleaf	4-6kg per hectare
Desmodium silverleaf	4-6kg per hectare
Leucaena	10-15kg per hectare

iv) Establishment of grass-legume pastures

Generally, a good seedbed should be prepared and the fertilizer broadcast and disced into the soil. Grasses and legumes can be established by methods outlined in earlier. Seed should have a shallow cover of soil, achieved either by rolling or light harrowing.

Alternatively, legumes may be over-sown into already established pastures. This may be done by either grazing down or burning the pasture (if fire- resistant) at the start of the rains and broadcasting the seed on top; or sowing in narrow strips that first have been ploughed or harrowed across the pasture.

Leucaena preferably should be established the year before the grass is planted seedlings are very sensitive to weed and grass competition.

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v) Fertilizer

In the establishment year a basal application of 300 kg "d" compound per hectare should be made and incorporated into the soil. Higher levels of nitrogen application should not be made as nodulation of the legumes may be reduced.

At the start of the rains in subsequent years, a top dressing of 200-300kg single supers per ha should be applied.

If the grass component of the mixture shows poor growth and signs of being dominated by the legume, then the grass may be encouraged by small applications of nitrogen (60kg urea per ha or 80kg ammonium nitrate per ha). These should be applied immediately after the pasture has been grazed. Large applications of nitrogen should be avoided as they will result in too lush a growth of grass and interfere with nitrogen fixation by the legume nodules.

vi) Management

Correct management is essential for maintenance of stable grass-legume mixture. This can be summaries as follows:

- Do not apply high levels of nitrogen;
- Apply annual dressings of phosphorous and sulphur (single super phosphate contain a minimum of 12 percent sulphur)
- Do not overgraze;
- Allow 6-8 weeks' rest period between grazings to permit the slower-growing legume to recover satisfactorily.

Star grass-legume mixture:

Since star grass will take a season or more to fully establish itself, legumes have time to grow and spread without much competition during the first 2-3 months. As soon as the star grass begins to thicken and over-grow the legume, the pasture should be slashed 10cm above ground level or lightly grazed. This has the dual purpose of encouraging lateral spread of the grass and preventing shading out of the legumes before adequately established. First grazings should be light to remove the grass canopy and to avoid grazing and trampling of young legume plants.

Rhodes/Buffel grass-legume mixture:

Under favourable conditions, Rhodes and Buffel grasses will establish and grow rapidly. To prevent shading out and depression of legume growth during its establishment, the grasses should be slashed or lightly grazed before the legume becomes over grown. As for star grass, all initial grazings should be light until the mixed sward is fully established.

Guinea grass-legume mixtures:

Establishment of Guinea is generally quite slow and there is not much lateral spread. If slashing or grazing is necessary to prevent legumes becoming over-grown during establishment, then the Guinea grass should not be cut below 15-20 cm from the ground.

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